

Spherical Hermite sums on modulation spaces

A complete classification in dimensions at least three

Codex agent lane 6

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Status: candidate partial result, likely valid. The source question is completely answered for every $d \geq 3$, but the intermediate range in $d = 2$ remains open. Human review is recommended.

Source question

Jaming and Speckbacher ask in Remark 3.3 of *Convergence of Hermite expansions in modulation spaces*, arXiv:2601.04893 [1], for which p the spherical Hermite sums

$$\mathring{S}_N f = \sum_{\substack{n \in \mathbb{N}_0^d \\ |n| \leq N}} \langle f, h_n \rangle h_n$$

converge in $M^p(\mathbb{R}^d)$. Here h_n are the tensor Hermite functions and

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P. JAMING AND M. SPECKBACHER

Proposition 3.2. *Let $1 \leq p < \infty$, $f \in M^p(\mathbb{R}^d)$ and $(x)_+ := x \cdot \chi_{[0, \infty)}(x)$. If $\alpha > (d-1)\left|\frac{1}{2} - \frac{1}{p}\right|$, then the Bochner-Riesz means*

$$B_R^\alpha f := \sum_{n \in \mathbb{N}_0^d} \left(1 - \frac{|n|^2}{R^2}\right)_+^\alpha \langle f, h_n \rangle h_n$$

converge to f in $M^p(\mathbb{R}^d)$ as $R \rightarrow \infty$.

Proof. Since the family of Fourier multipliers A_{m_R} with $m_R(n) = (1 - |n|^2/R^2)_+^\alpha$ is uniformly bounded on $L^p(\mathbb{T}^d)$ whenever $\alpha > (d-1)\left|\frac{1}{2} - \frac{1}{p}\right|$ (see, e.g., [11] Proposition 4.1.9)), the result follows immediately from Proposition 2.2. $\square$

Remark 3.3. Since [3] Proposition 4.3] provides only a sufficient condition for the boundedness of Hermite multipliers, it remains an open problem whether certain symbols that generate unbounded Fourier multipliers also yield unbounded Hermite multipliers. A natural example to consider is Fefferman's famous counterexample for the ball multiplier [5]. This leads to the following question: for which range of p does the sequence of spherical partial sums in the Hermite expansion,

$$\mathring{S}_N f := \sum_{n \in \mathbb{N}_0^d, |n| \leq N} \langle f, h_n \rangle h_n$$

converge in $M^p(\mathbb{R}^d)$?

APPENDIX A. UPPER FRAME AND BESSEL BOUNDS OF GABOR SYSTEMS
ASSOCIATED TO HERMITE FUNCTIONS

The image is a full-width crop of source page 10. The question follows the authors' observation that the known implication from torus Fourier multipliers to Hermite multipliers is only sufficient; they specifically point to Fefferman's ball multiplier as the natural obstruction.

Result and scope

Theorem 1 (High-dimensional classification). *Let $d \geq 3$ and $1 \leq p \leq \infty$. The following are equivalent:*

1. *for every $f \in M^p(\mathbb{R}^d)$, one has $\dot{S}_N f \rightarrow f$ in $M^p(\mathbb{R}^d)$;*
2. *$p = 2$.*

In particular, for every $p \neq 2$,

$$\sup_{N \in \mathbb{N}} \|\dot{S}_N\|_{M^p \rightarrow M^p} = \infty.$$

For $d = 2$, convergence holds at $p = 2$ and fails at $p = 1, \infty$; the present proof does not decide $1 < p < \infty, p \neq 2$.

Proof intuition

The Bargmann transform sends Hermite functions to normalized monomials. A finite collection of monomials whose multi-indices lie near a very large base index K has, after division by the norm of the base monomial, the same asymptotic norm as the corresponding trigonometric polynomial on a torus. This is an elementary concentration fact: the radial variables under the mass of a large monomial are Gamma distributed and concentrate at the base index.

We place the neighboring indices in the integer hyperplane $v^\perp$. Pell's equation produces arbitrarily remote base indices K and integer radii N for which the spherical condition is *exactly* $|q| \leq J$ on that hyperplane. Thus a hypothetical uniform bound for spherical Hermite sums would imply uniform bounds for ellipsoidal Fourier sums on a $(d-1)$ -torus. When $d \geq 3$, de Leeuw transference and Fefferman's theorem make this impossible unless $p = 2$.

Bargmann-Fock setup

Under the standard Bargmann transform, $M^p(\mathbb{R}^d)$ is isomorphic (isometric up to the normalization convention) to the Fock space

$$\mathcal{F}^p(\mathbb{C}^d) = \left\{ F \text{ entire} : \int_{\mathbb{C}^d} |F(z)|^p e^{-\pi p |z|^2/2} dz < \infty \right\}, \quad 1 \leq p < \infty,$$

and h_n is sent to

$$e_n(z) = \frac{\pi^{|n|/2}}{\sqrt{n!}} z^n, \quad n \in \mathbb{N}_0^d.$$

Consequently $\dot{S}_N$ becomes the monomial multiplier with symbol $1_{\{|n| \leq N\}}$.

Lemma 1 (High-monomial torus limit). *Fix $1 \leq p < \infty$, a vector $v \in \mathbb{N}^d$ all of whose coordinates are positive, a finite set*

$$E \subset H_v := \{q \in \mathbb{Z}^d : v \cdot q = 0\},$$

and scalars c_q . For $K = Lv$, with L sufficiently large that $K + q \in \mathbb{N}_0^d$ for every $q \in E$, put

$$F_L = \sum_{q \in E} c_q e_{K+q}.$$

Then

$$\lim_{L \rightarrow \infty} \frac{\|F_L\|_{\mathcal{F}^p}}{\|e_K\|_{\mathcal{F}^p}} = \left\| \sum_{q \in E} c_q e^{iq \cdot \theta} \right\|_{L^p(\mathbb{T}^d)},$$

where Haar measure on the torus is normalized.

Proof. Write $z_j = r_j e^{i\theta_j}$ and $t_j = \pi r_j^2$. Normalize the measure

$$|e_K(z)|^p e^{-\pi p |z|^2/2} dz$$

to have mass one. Its angular part is normalized Haar measure, and its radial variables are independent with

$$t_j \sim \text{Gamma}\left(\frac{pK_j}{2} + 1, \text{rate } \frac{p}{2}\right).$$

Furthermore,

$$\frac{e_{K+q}(z)}{e_K(z)} = A_{K,q}(t) e^{iq \cdot \theta}, \quad A_{K,q}(t) = \prod_{j=1}^d t_j^{q_j/2} \sqrt{\frac{K_j!}{(K_j + q_j)!}}.$$

For fixed q , $t_j/K_j \rightarrow 1$ in probability. The factorial ratio asymptotic therefore gives $A_{K,q} \rightarrow 1$ in probability. This convergence also holds in every fixed finite L^s . Indeed, for all sufficiently large K_j , the Gamma moment formula gives

$$\begin{aligned} \mathbb{E}[A_{K,q}^s] &= \prod_{j=1}^d \left(\frac{K_j!}{(K_j + q_j)!} \right)^{s/2} \left(\frac{p}{2} \right)^{-sq_j/2} \\ &\quad \times \frac{\Gamma(pK_j/2 + 1 + sq_j/2)}{\Gamma(pK_j/2 + 1)} \rightarrow 1. \end{aligned}$$

Applying this with a moment strictly larger than p supplies uniform integrability, so $A_{K,q} \rightarrow 1$ in L^p . Since E is finite, Minkowski's inequality yields

$$\left\| \sum_{q \in E} c_q (A_{K,q} - 1) e^{iq \cdot \theta} \right\|_{L^p} \leq \sum_{q \in E} |c_q| \|A_{K,q} - 1\|_{L^p} \rightarrow 0.$$

This is exactly the asserted normalized norm limit. $\square$

Exact integer-radius placement

Lemma 2 (Pell slice). *For every $d \geq 3$, there is a primitive vector $v \in \mathbb{N}^d$, all of whose coordinates are positive, for which $D = |v|^2$ is not a square. For each integer $J \geq 1$, there are sequences $K_k \in \mathbb{N}^d$ and $N_k \in \mathbb{N}$, with $\min_j (K_k)_j \rightarrow \infty$, such that for every $q \in H_v$ and all large k for which $K_k + q \geq 0$,*

$$|K_k + q| \leq N_k \iff |q| \leq J.$$

Proof. If d is not a square, take $v = (1, \dots, 1)$, so $D = d$. If d is a square, take $v = (1, \dots, 1, 2)$, with $d - 1$ entries equal to one; then $D = d + 3$. For square $d \geq 4$, $d + 3$ cannot be a square, because two positive squares differ by three only in the pair 1, 4. In either case v is primitive and D is nonsquare.

Pell's equation

$$a^2 - Db^2 = 1$$

has positive solutions (a_k, b_k) with $b_k \rightarrow \infty$. Set

$$K_k = Jb_kv, \quad N_k = Ja_k.$$

For $q \in H_v$, orthogonality and Pell's identity give

$$|K_k + q|^2 = DJ^2b_k^2 + |q|^2 = N_k^2 - J^2 + |q|^2.$$

The desired equivalence follows, with no rounding or limiting error. $\square$

Proof of the theorem

Let first $1 < p < \infty$, and suppose that $\mathring{S}_N f \rightarrow f$ in M^p for every f . Banach–Steinhaus yields

$$\sup_N \|\mathring{S}_N\|_{M^p \rightarrow M^p} = C < \infty.$$

Transfer this bound to $\mathcal{F}^p(\mathbb{C}^d)$. Choose v from Lemma 2. Since v is primitive, the lattice H_v is saturated, has rank $d - 1$, and is the character lattice of a $(d - 1)$ -torus.

Fix $J \geq 1$ and a finite trigonometric polynomial

$$P(\theta) = \sum_{q \in E} c_q e^{iq \cdot \theta}, \quad E \subset H_v.$$

Embed it as $F_k = \sum_{q \in E} c_q e_{K_k+q}$, where K_k, N_k come from Lemma 2. The exact Pell equivalence gives

$$\mathring{S}_{N_k} F_k = \sum_{\substack{q \in E \\ |q| \leq J}} c_q e_{K_k+q}.$$

Apply the uniform Fock bound, divide by $\|e_{K_k}\|_{\mathcal{F}^p}$, and use Lemma 1 on both sides. We obtain

$$\left\| \sum_{\substack{q \in E \\ |q| \leq J}} c_q e^{iq \cdot \theta} \right\|_p \leq C \left\| \sum_{q \in E} c_q e^{iq \cdot \theta} \right\|_p. \quad (1)$$

Choose a lattice isomorphism $B : \mathbb{Z}^{d-1} \rightarrow H_v$. In these coordinates, (1) says that the Fourier multipliers

$$m_J(\ell) = 1_{\{Q(\ell) \leq J^2\}}, \quad Q(\ell) = |B\ell|^2,$$

are uniformly bounded on $L^p(\mathbb{T}^{d-1})$. The dilation part of de Leeuw's multiplier theorem [2] then makes the ellipsoid indicator $1_{\{Q(\xi) \leq 1\}}$ an $L^p(\mathbb{R}^{d-1})$ Fourier multiplier. The ellipsoid boundary has measure zero, so the usual a.e.-continuity formulation applies. After an invertible linear change of variables, this is the Euclidean ball multiplier. Since $d - 1 \geq 2$, Fefferman's theorem [3] forces $p = 2$, a contradiction whenever $p \neq 2$.

At $p = 1$, take $f = g \otimes h_0^{\otimes(d-1)}$. Tensor factorization of the modulation norm and the Hermite coefficients gives

$$\mathring{S}_N f = (S_N g) \otimes h_0^{\otimes(d-1)},$$

where S_N is the one-dimensional Hermite partial sum. The source proves that $\|S_N\|_{M^1(\mathbb{R}) \rightarrow M^1(\mathbb{R})}$ grows at least logarithmically [1, Theorem 1.1]; hence the spherical sums are not uniformly bounded on $M^1(\mathbb{R}^d)$. Under the standard M^1 – M^∞ duality, these finite-rank real diagonal projections are adjoints of themselves, so their M^∞ operator norms equal their M^1 norms. This rules out $p = \infty$.

Finally, $M^2(\mathbb{R}^d) = L^2(\mathbb{R}^d)$ with equivalent (indeed, conventionally isometric) norm by Moyal's identity. The operators $\mathring{S}_N$ are increasing orthogonal projections associated with the tensor Hermite orthonormal basis, so $\mathring{S}_N f \rightarrow f$ for every $f \in M^2$. This proves the theorem.

Dimension two, verification, and limitations

The proof above exposes the precise dimension threshold: H_v has rank $d - 1$. When $d = 2$, it is one-dimensional, and its ball multipliers are interval partial sums, uniformly bounded for $1 < p < \infty$. Thus this method does not settle the intermediate two-dimensional range. The tensor endpoint argument and the $p = 2$ argument remain valid, giving the scoped claims stated in the theorem.

The included script checks Pell identities for representative dimensions and radii and checks the convergence of the exact Gamma moments for positive and negative shifts. These computations are sanity checks, not parts of the proof.

A bounded novelty search on August 11, 2026 covered the local run indexes, the exact source title and question phrases, Hermite/Fock multiplier converses, monomial ball sums in Fock spaces, and Pell-based multiplier transference. It found the source and general multiplier literature, but no later answer to Remark 3.3 and no matching argument. This is not an exhaustive novelty claim.

Human-review recommendation. Check Lemma 1's normalization and the discontinuous-symbol form of de Leeuw dilation. The Pell placement is an exact integer identity. The packet must remain classified as partial until the intermediate ($d=2$) range is settled.

References

- [1] P. Jaming and M. Speckbacher, *Convergence of Hermite expansions in modulation spaces*, arXiv:2601.04893, 2026.
- [2] K. de Leeuw, *On L_p multipliers*, Ann. of Math. (2) **81** (1965), 364–379.
- [3] C. Fefferman, *The multiplier problem for the ball*, Ann. of Math. (2) **94** (1971), 330–336.